07_Assignment_Yuliia_Chernysheva

October 11, 2026

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October 11, 2026
CSC 2267

1 Data Source Attribution and Citation

Source data and content for this assignment is "Student projects for Murach's Python for Data Analysis" made available by the publisher for courses using this textbook.

2 Guidelines

- Start your Notebook by importing any modules, classes, or functions required by the project.
- Use one code cell to store the code that accomplishes each task and question.
- Use comments to describe your thought process and to document any code that's hard to understand.
- Use Markdown cells to create headings that organize your Notebooks.

3 Dataset Summary

This dataset contains historical data on avocado prices and sales volume in multiple U.S. markets.

One of the columns in this dataset, `Unnamed: 0`, contains sequential numbers that are irrelevant to analyzing this data.

Three of the other columns contain sales for PLU (price look-up) codes `4046`, `4225`, and `4770`. These columns will also not be used by the analyses done in these projects.

If you review the data, you'll see that some of the regions overlap. For example, one of the regions is the entire U.S., and all of the other regions are parts of the U.S. Because of that, you would need to review this data carefully before determining the best way to analyze it. For the purposes of these exercises, though, the overlapping regions won't be taken into consideration.

[1]:
```
# Import required modules and libraries
import pandas as pd
```

[2]:
```
# Read the data from the CSV file into a DataFrame
# and inspect the initial records
```

```
df_avocado = pd.read_csv("avocado.csv")
print("=> Initial five rows of the avocado dataset:\n")
display(df_avocado.head())

# Compute the number of missing values for each column
# to ensure data integrity
print("\n=> Missing values summary by column:\n")
print(df_avocado.isnull().sum())

# Extract and inspect unique geographical regions
print("\n=> Frequency count of unique regions:\n")
display(df_avocado.region.value_counts())

# Drop rows that have a region of TotalUS
# and compute the number of dropped rows
total_count = df_avocado.shape[0]
df_avocado = df_avocado[df_avocado.region != 'TotalUS']
print("\n=> Total rows dropped after removing nationwide aggregates:",
      total_count - df_avocado.shape[0])

# Drop the Unnamed: 0, 4046, 4225, and 4770 columns
# that are not required for analysis
columns_to_drop = ['Unnamed: 0', '4046', '4225', '4770']
df_avocado.drop(columns=columns_to_drop, inplace=True)

# Convert the Date column to the DateTime type.
df_avocado.Date = pd.to_datetime(df_avocado.Date)

# Standardize column headers by removing spaces and applying title case␣
 ↪formatting
df_avocado.columns = df_avocado.columns.str.replace(" ","").str.strip()
df_avocado.columns = [col[0].upper() + col[1:] for col in df_avocado.columns]

# Calculate the proportion of extra-large bags relative to total bag volume
df_avocado['PctXLargeBags'] = (df_avocado.XLargeBags / df_avocado.TotalBags) *␣
 ↪100

# Calculate the proportion of large bags relative to total bag volume.
df_avocado['PctLargeBags'] = (df_avocado.LargeBags / df_avocado.TotalBags) * 100

# Calculate the proportion of small bags relative to total bag volume
df_avocado['PctSmallBags'] = (df_avocado.SmallBags / df_avocado.TotalBags) * 100

# Display the first five rows of data to view the new columns
print("\n=> Updated dataset featuring calculated bag size percentages:\n")
display(df_avocado.head())
```

```python
# Extract a subset dataframe containing only the target dimensions for␣
 ↪aggregation
df_avocado_new = df_avocado[['Region', 'Type', 'Year', 'TotalBags']].copy()

# Aggregate total bag volume by region, year, and type,
# then pivot avocado types into separate columns
print("\n=> Final aggregated and unstacked summary table:\n")
df_avocado_new.groupby(['Region', 'Year', 'Type']).sum().unstack(level='Type')
```

```
=> Initial five rows of the avocado dataset:

   Unnamed: 0        Date  AveragePrice  Total Volume     4046       4225  \
0           0  2015-12-27          1.33      64236.62  1036.74   54454.85
1           1  2015-12-20          1.35      54876.98   674.28   44638.81
2           2  2015-12-13          0.93     118220.22   794.70  109149.67
3           3  2015-12-06          1.08      78992.15  1132.00   71976.41
4           4  2015-11-29          1.28      51039.60   941.48   43838.39

     4770  Total Bags  Small Bags  Large Bags  XLarge Bags          type  \
0   48.16     8696.87     8603.62       93.25          0.0  conventional
1   58.33     9505.56     9408.07       97.49          0.0  conventional
2  130.50     8145.35     8042.21      103.14          0.0  conventional
3   72.58     5811.16     5677.40      133.76          0.0  conventional
4   75.78     6183.95     5986.26      197.69          0.0  conventional

   year  region
0  2015  Albany
1  2015  Albany
2  2015  Albany
3  2015  Albany
4  2015  Albany


=> Missing values summary by column:

Unnamed: 0      0
Date            0
AveragePrice    0
Total Volume    0
4046            0
4225            0
4770            0
Total Bags      0
Small Bags      0
Large Bags      0
XLarge Bags     0
type            0
```

```
year            0
region          0
dtype: int64

=> Frequency count of unique regions:


region
Albany                338
Atlanta               338
BaltimoreWashington   338
Boise                 338
Boston                338
BuffaloRochester      338
California            338
Charlotte             338
Chicago               338
CincinnatiDayton      338
Columbus              338
DallasFtWorth         338
Denver                338
Detroit               338
GrandRapids           338
GreatLakes            338
HarrisburgScranton    338
HartfordSpringfield   338
Houston               338
Indianapolis          338
Jacksonville          338
LasVegas              338
LosAngeles            338
Louisville            338
MiamiFtLauderdale     338
Midsouth              338
Nashville             338
NewOrleansMobile      338
NewYork               338
Northeast             338
NorthernNewEngland    338
Orlando               338
Philadelphia          338
PhoenixTucson         338
Pittsburgh            338
Plains                338
Portland              338
RaleighGreensboro     338
RichmondNorfolk       338
Roanoke               338
```

```
Sacramento              338
SanDiego                338
SanFrancisco            338
Seattle                 338
SouthCarolina           338
SouthCentral            338
Southeast               338
Spokane                 338
StLouis                 338
Syracuse                338
Tampa                   338
TotalUS                 338
West                    338
WestTexNewMexico        335
Name: count, dtype: int64

=> Total rows dropped after removing nationwide aggregates: 338

=> Updated dataset featuring calculated bag size percentages:

        Date  AveragePrice  TotalVolume  TotalBags  SmallBags  LargeBags  \
0 2015-12-27          1.33     64236.62    8696.87    8603.62      93.25
1 2015-12-20          1.35     54876.98    9505.56    9408.07      97.49
2 2015-12-13          0.93    118220.22    8145.35    8042.21     103.14
3 2015-12-06          1.08     78992.15    5811.16    5677.40     133.76
4 2015-11-29          1.28     51039.60    6183.95    5986.26     197.69

   XLargeBags          Type  Year  Region  PctXLargeBags  PctLargeBags  \
0         0.0  conventional  2015  Albany            0.0      1.072225
1         0.0  conventional  2015  Albany            0.0      1.025610
2         0.0  conventional  2015  Albany            0.0      1.266244
3         0.0  conventional  2015  Albany            0.0      2.301778
4         0.0  conventional  2015  Albany            0.0      3.196824

   PctSmallBags
0     98.927775
1     98.974390
2     98.733756
3     97.698222
4     96.803176

=> Final aggregated and unstacked summary table:
```

[2]:

	TotalBags	
Type	conventional	organic

```
Region           Year
Albany           2015    662366.17    57289.43
                 2016    759091.14    79208.77
                 2017    699561.17   135944.43
                 2018    245240.75    41552.75
Atlanta          2015   2935925.63    61065.24
…                              …           …
West             2018  35978608.70  2074684.34
WestTexNewMexico 2015   5399316.40   128763.94
                 2016  10870856.07   403881.08
                 2017  14195031.44   602152.24
                 2018   3547372.46   149141.90

[212 rows x 2 columns]
```

[ ]: